



COMMUNITY EVENT #2

AI for Health

SPEAKER



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PACSOI

PAtient-Centric SOld



FAQIR
INSTITUTE

FAQIR
FOUNDATION

IDLab·umec
INTERNET & DATA LAB



 **Byteflies**

 **moveUP**

A CONTRARIO

 **VLAIO**



An avalanche of health data is available
... but **hard to use, not interoperable**

The **patient wants to be involved** in his/her own health



Data is stored in silos, unstructured, not linkable ... so **difficult to find, access, integrate or (re)use**

Challenge to deal with **data generated by the patient** outside 'the system'.



Health data standards exist but need to be used and combined

Innovative or personalised therapies require **integrated patient data**



Separate data from applications



From Fragmented Data



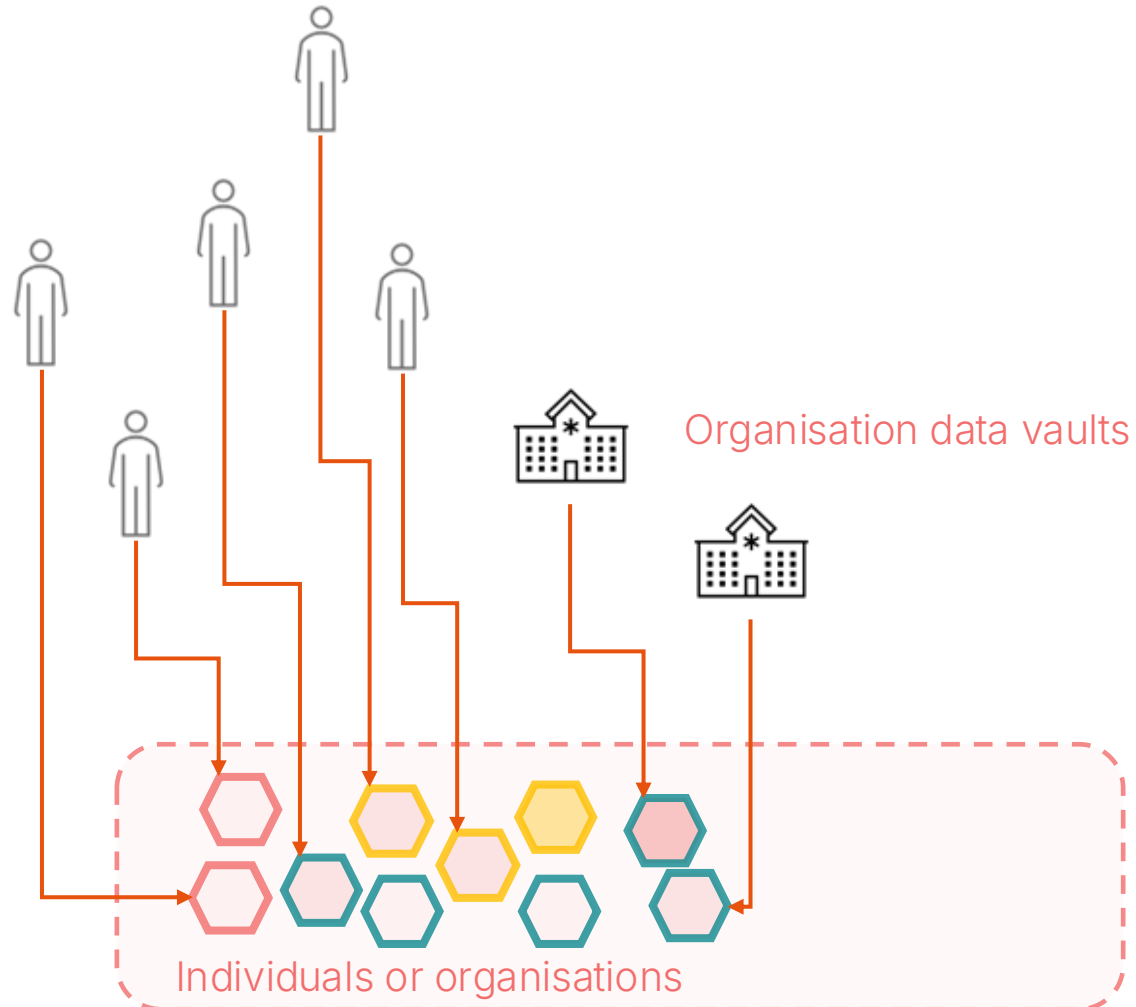
to Personal Data Vaults





PACSOI Ecosystem

Person-centric data vaults



FAQIR
FOUNDATION

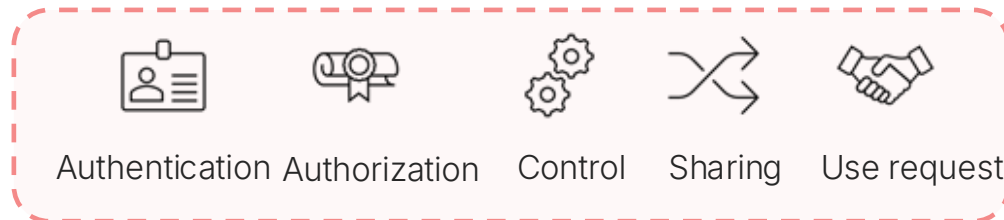
IDLab ^{umec}
INTERNET & DATA LAB

AI
TECH HUB

PACSOI Ecosystem



Tools, policies, guidelines, audit logging for Data Vaults



A CONTRARIO



PACSOI Ecosystem



Person-Centric Standardized Data





PACSOI Ecosystem



Data exchange and connectivity tools





PACSOI Ecosystem

Healthcare, Research,
Industry, Government

Person-Centric Standardized Data

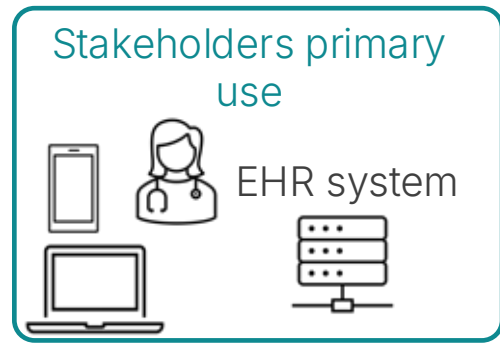
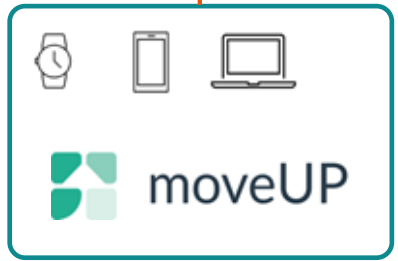
Healthcare system

Stakeholders primary
use

Stakeholders Secondary
use

Reports,
Analytics,
Registries

EHR system



Authentication Authorization Control Sharing Use request

ACONTRARIO



Individuals or organisations



Aggregators



The obesity use case

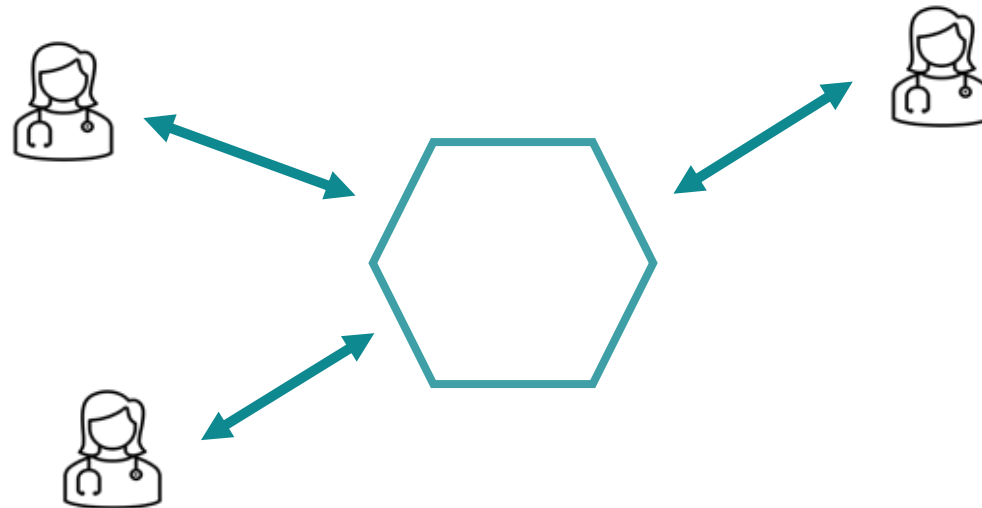
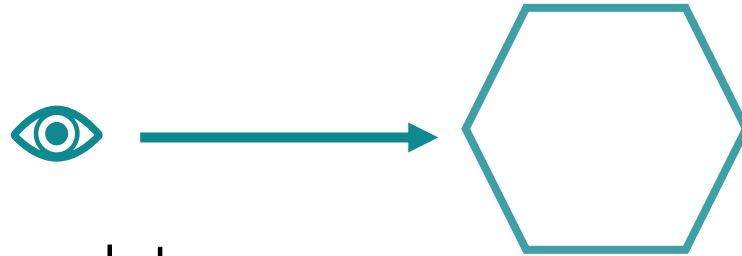
The first step

- Patient with a personal health history ...
 - Previous and existing conditions
 - Clinical examinations
 - Used medications
 - Previous efforts to loose weight
 - ...
- ... takes the decision to go for a ‘final’ solution
 - Bariatric surgery



The obesity use case Assessment

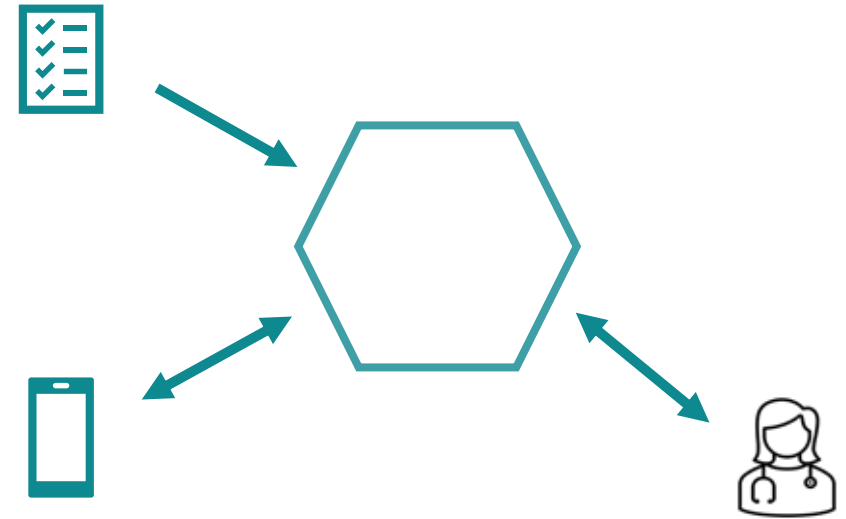
- Patient – Surgeon encounter
 - Patient eligible for surgery?
 - Share existing data – generate new data
 - Search for proof of prolonged obesity
 - Involve multidisciplinary team > share data with other HCPs
 - Endocrinologist
 - Dietician
 - Psychologist
 - ...



The obesity use case

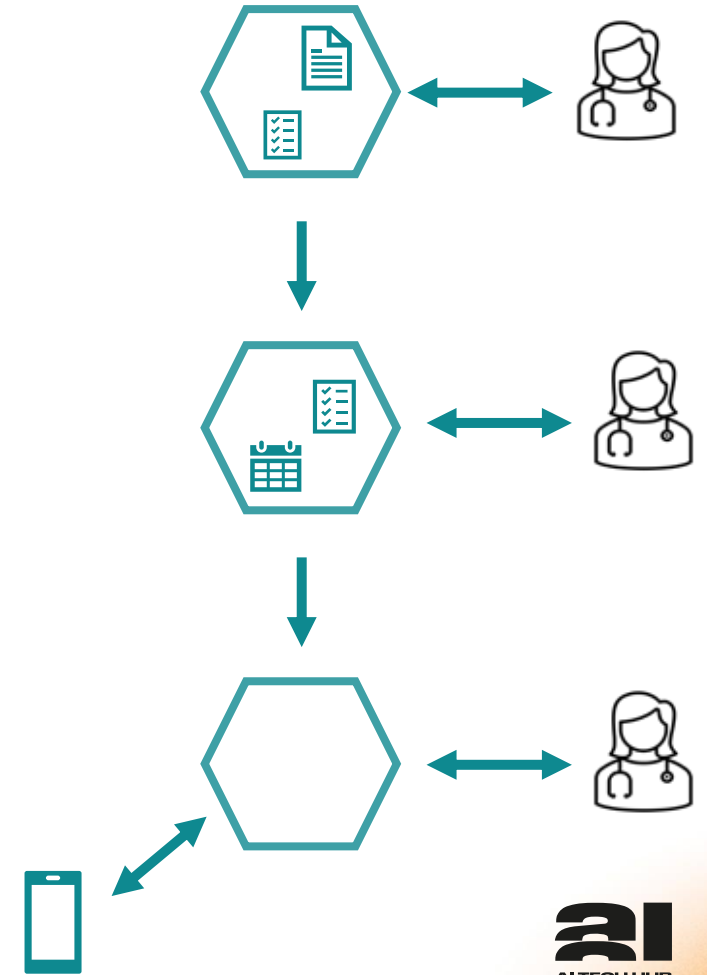
Prepare

- Start patient monitoring & follow-up
 - Patient ready for lifestyle change?
 - Prepare for surgery
 - Patient-generated data
 - Supported by digital health tools



The obesity use case Treatment & Recovery

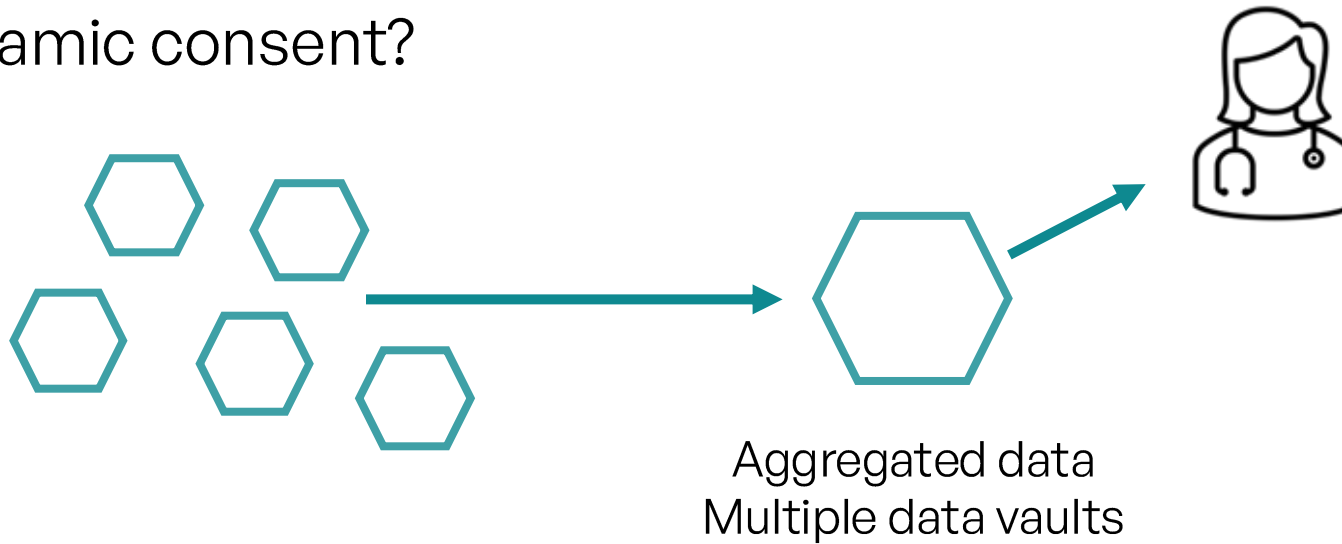
- Bariatric surgery
 - Hospital admission > data sharing with care team
 - Surgery
 - Post-operative care in the hospital
- Recovery at home
 - Guided follow-up using digital tooling
 - Monitor weight loss
- Follow-up until 2y after surgery



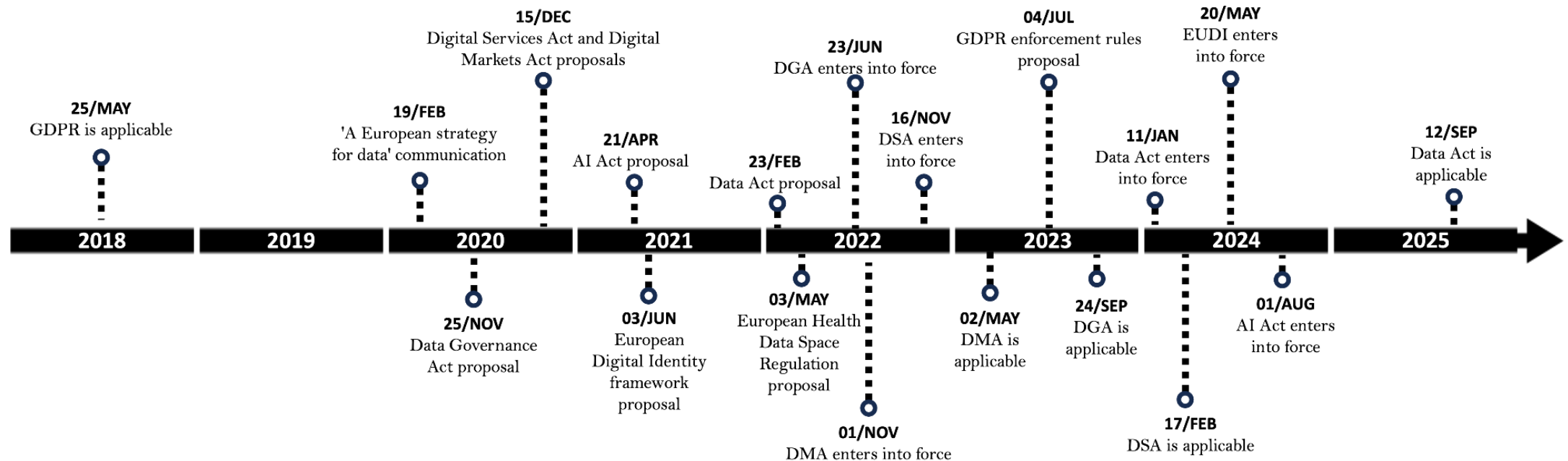
The obesity use case

Longitudinal follow-up

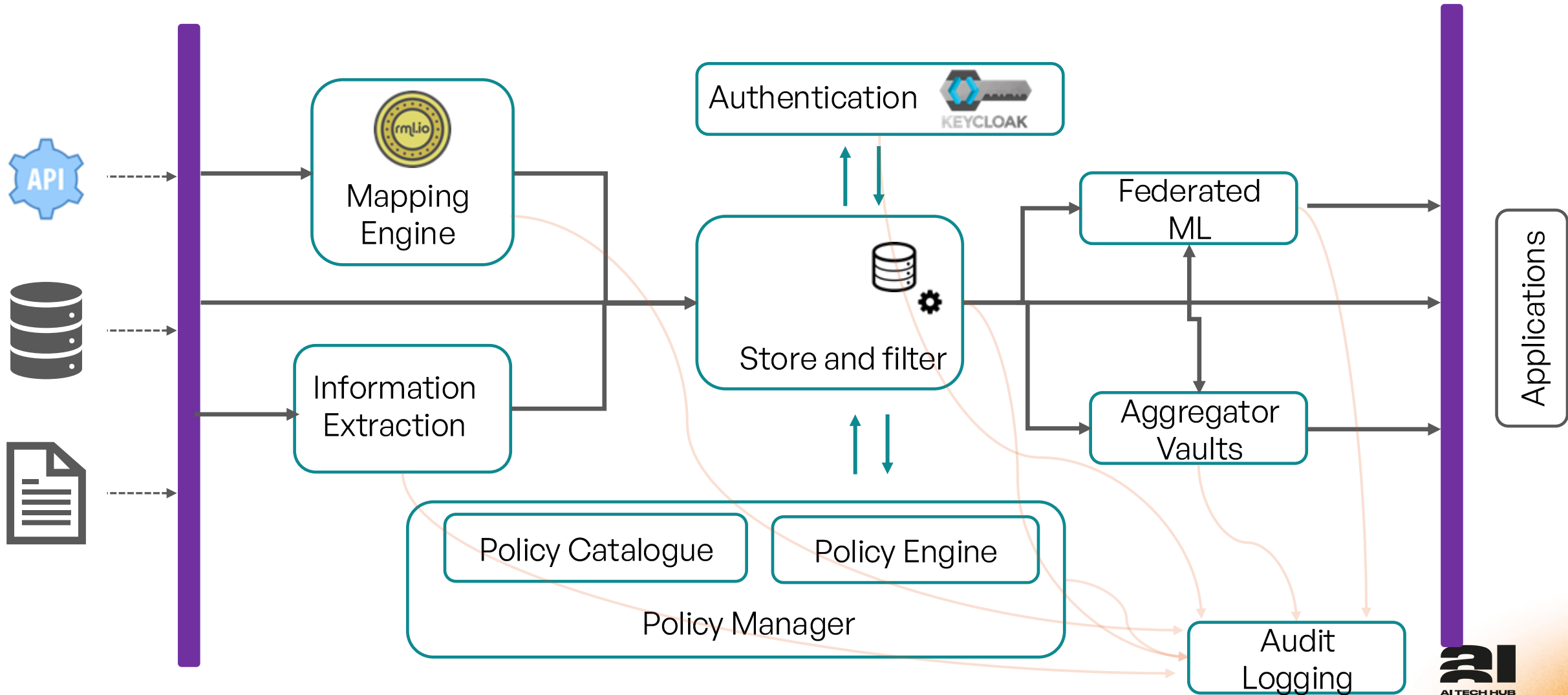
- No active patient – physician relation
- Secondary use of health data
 - Dynamic consent?



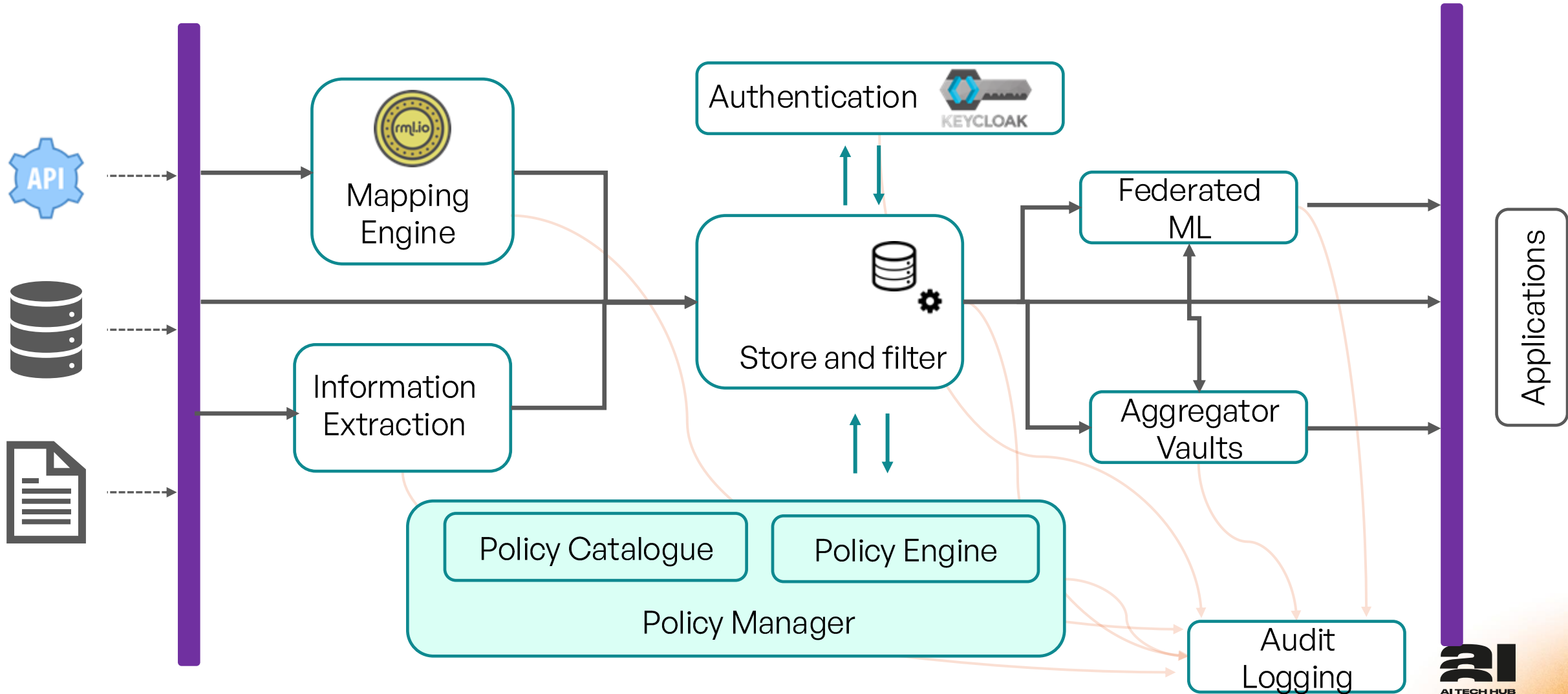
Data Protection and AI Laws Landscape



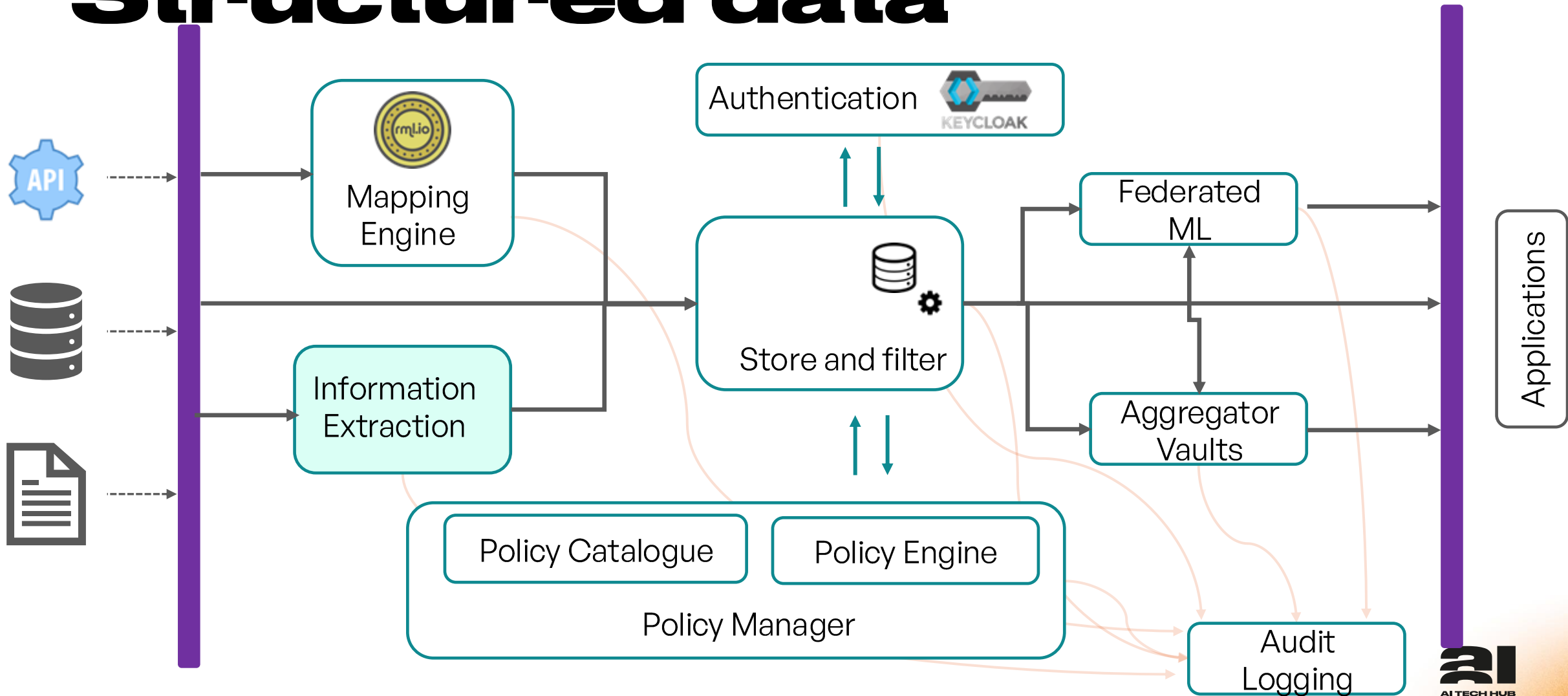
Architecture



Policy Management



Unstructured to Structured data



Policy Management



Involved entities



Involved action(s)



**Involved
resource(s)**



**Time
Location
Purpose
Legal ground**

Health Data Sharing



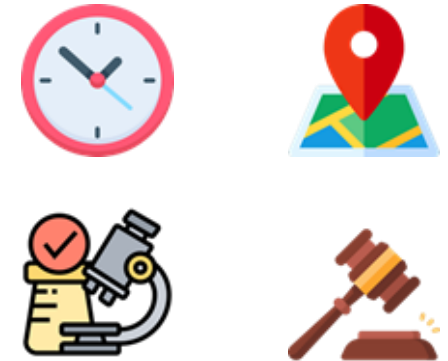
My GP



Read



**Blood Test
Results**



**During
consultation;
for clinical
diagnosis**

Who **[can|cannot|must]** act what in which resource how

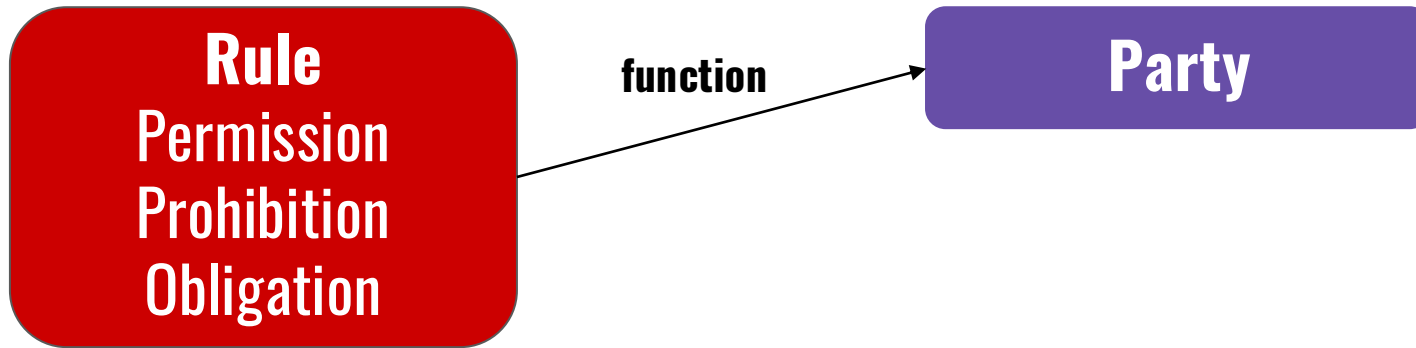
Rule

Permission

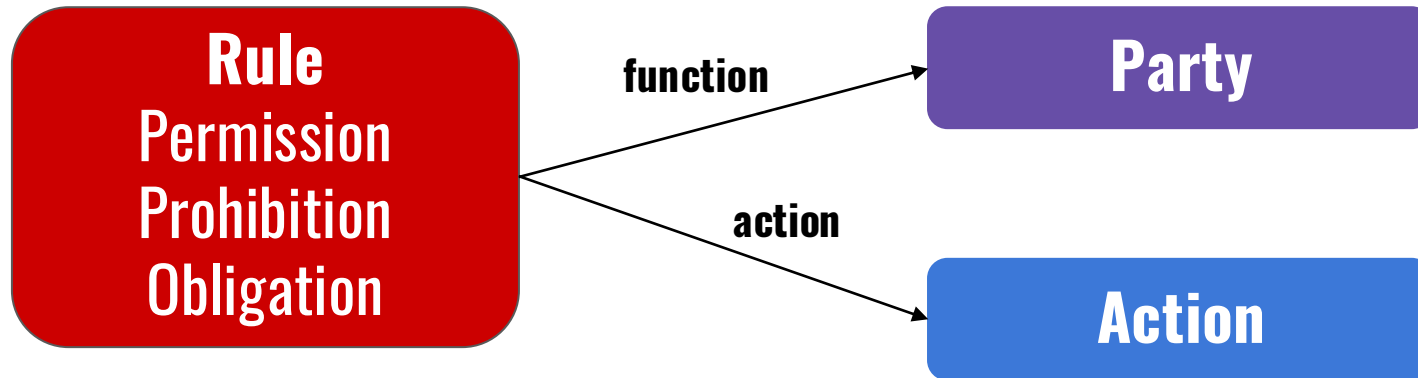
Prohibition

Obligation

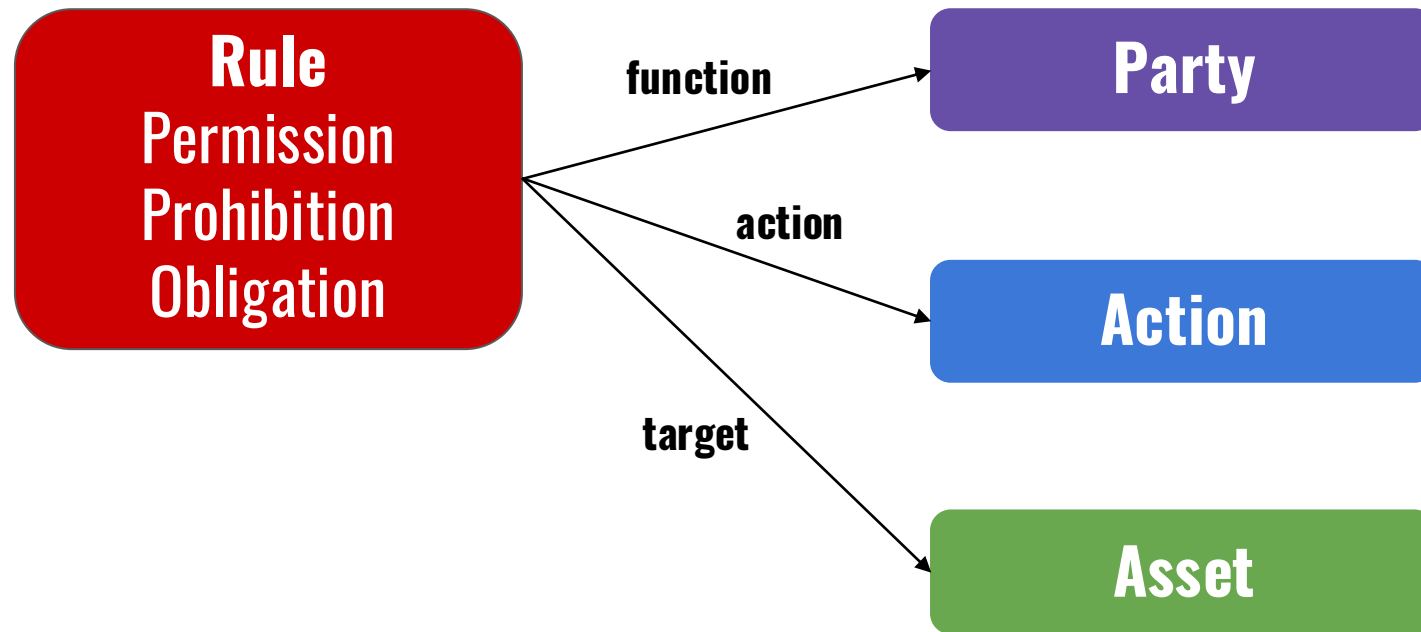
Who [can|cannot|must] act what in which resource how



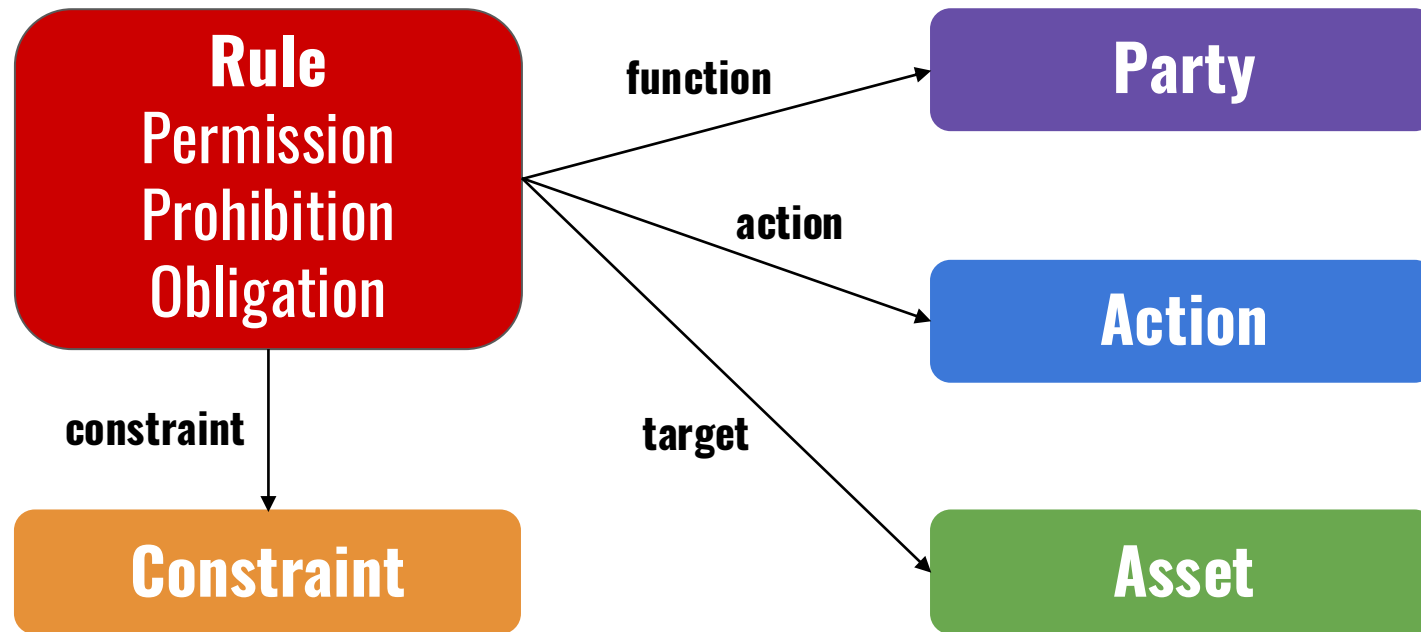
Who **[can|cannot|must]** act what in which resource how



Who [**can|cannot|must**] **act**
what **in which resource** **how**

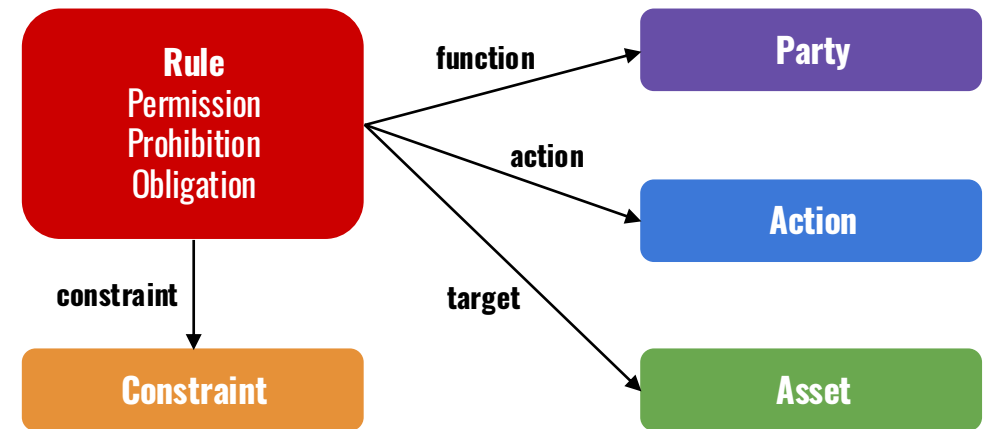


Who [can|cannot|must] act what in which resource how



Open Digital Rights Language (ODRL)

- Standard for the expression of policies
 - ODRL Information Model – W3C Recommendation
 - ODRL Core Vocabulary – W3C Recommendation
- Maintained by the W3C ODRL Community Group
- Composed by several other specifications
 - ODRL Implementation Best Practices
 - ODRL Profile Best Practices
 - ODRL Formal Semantics
 - ODRL Community Vocabulary
- Domain- and jurisdiction-agnostic
 - Easily extendable through the use of ODRL profiles

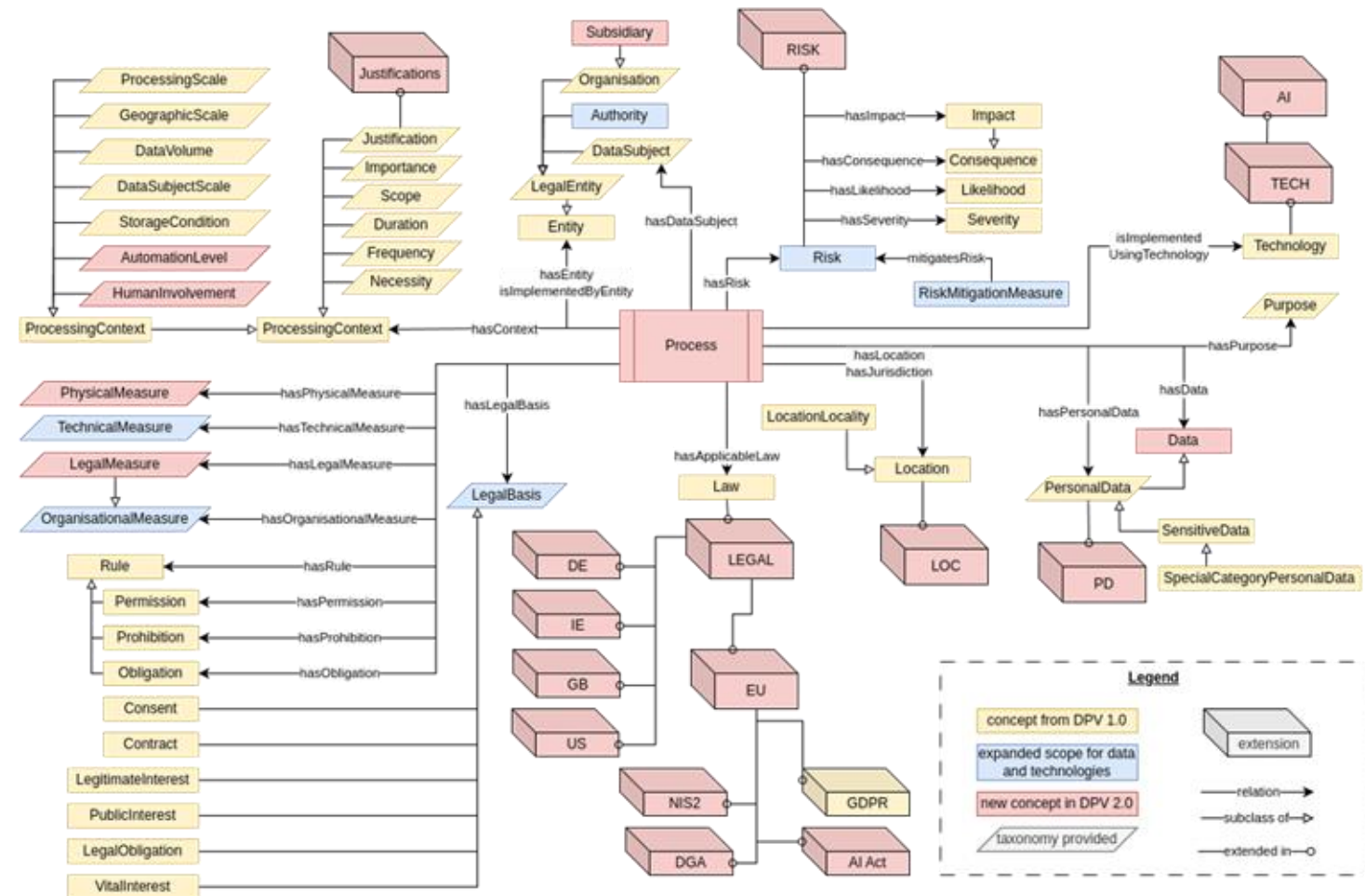


- Developed by the **W3C** Data Privacy Vocabularies and Controls Community Group (**DPVCG**)
- Defines a **jurisdiction-agnostic** ontology for expressing metadata about the processing of personal data
- Provides **hierarchical taxonomies**, from abstract to more specific concepts, to instantiate specific concepts in practical use-cases
- Has **law-specific extensions**

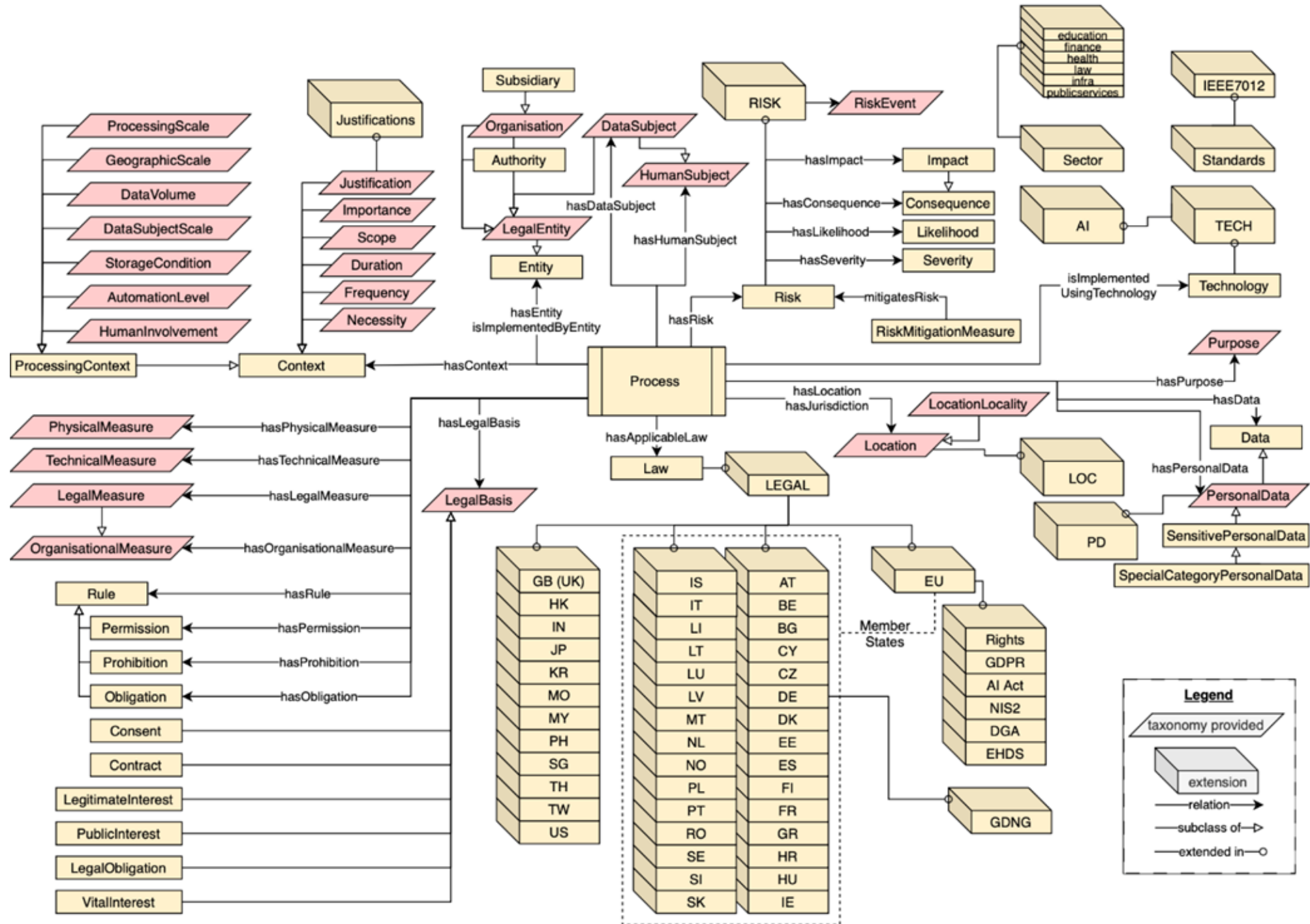
<https://w3id.org/dpv>

<https://w3id.org/dpv/primer>

Data Privacy Vocabulary (DPV)



Harshvardhan J. Pandit, Beatriz Esteves, Georg P. Krog, Paul Ryan, Delaram Golpayegani, and Julian Flake (2024). **Data Privacy Vocabulary (DPV) – Version 2.0**. *The Semantic Web – ISWC 2024*, pp. 171–93. doi: 10.1007/978-3-031-77847-6_10.

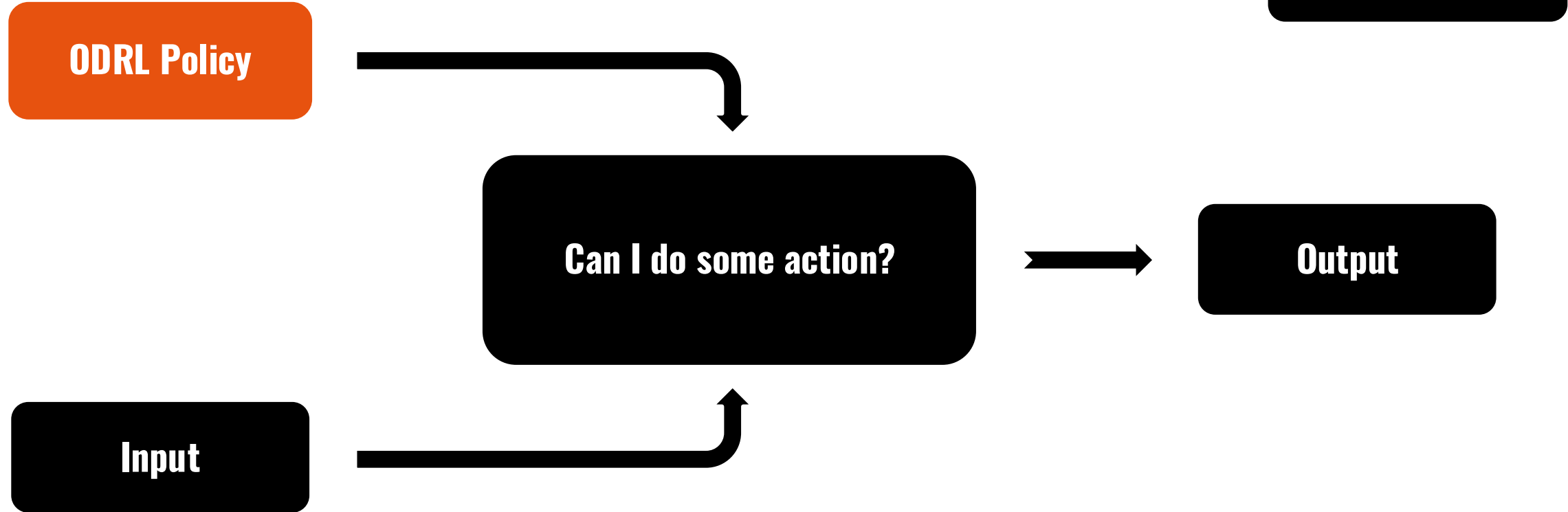


Legally-aligned Policies

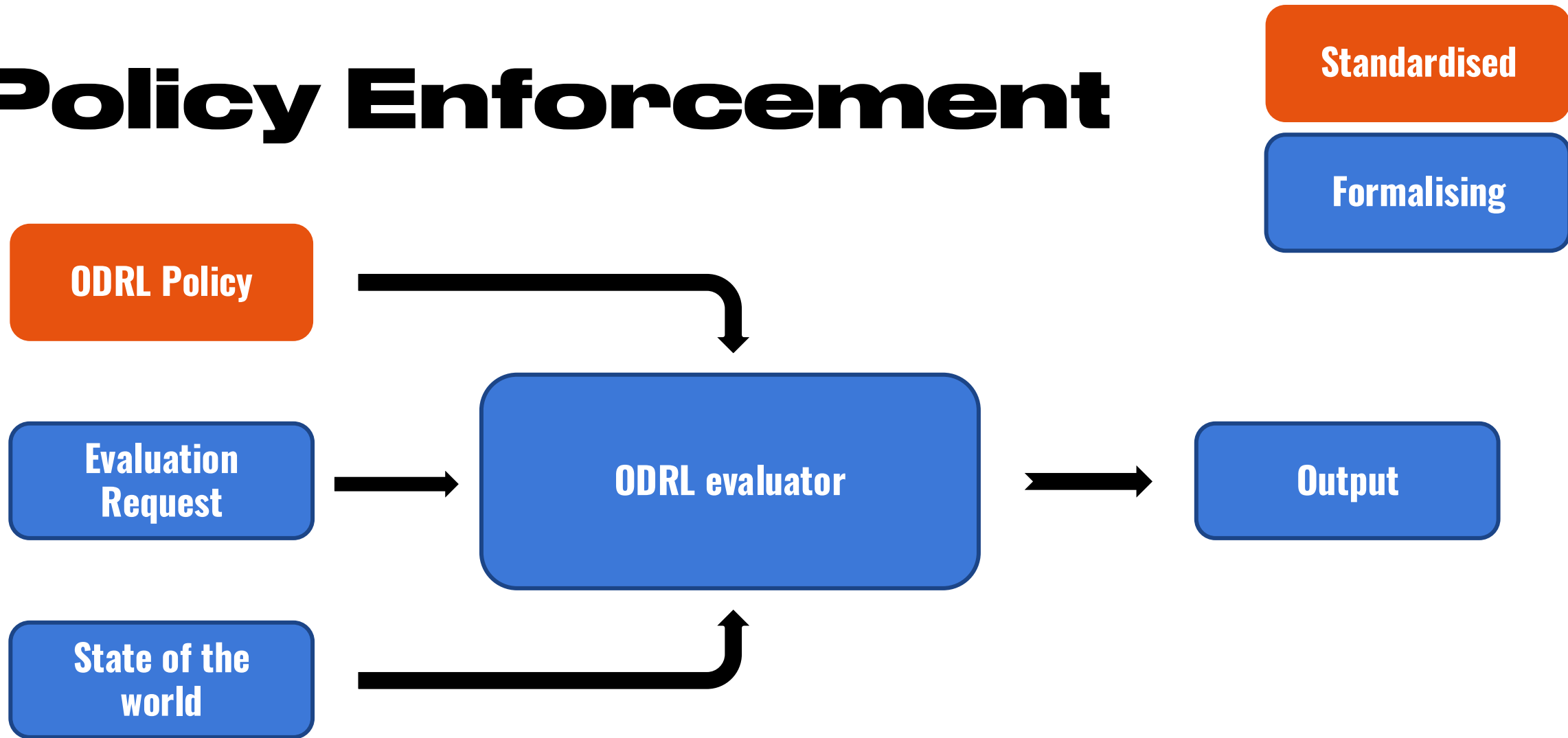
```
ex:policy-partycollection a odrl:Agreement ;
  odrl:uid ex:policy-partycollection ;
  odrl:profile dpv-odrl;;
  odrl:permission [
    odrl:assigner ex:data-holder ;
    odrl:assignee [
      a odrl:PartyCollection ;
      odrl:source ex:entities ;
      odrl:refinement [
        odrl:leftOperand dpv-odrl:Entity ;
        odrl:operator odrl:isA ;
        odrl:rightOperand dpv:AcademicScientificOrganisation ] ] ;
    odrl:target ex:research-dataset ;
    odrl:action dpv-odrl:Use ;
    odrl:constraint [
      odrl:leftOperand dpv-odrl:Purpose ;
      odrl:operator odrl:isA ;
      odrl:rightOperand dpv:ResearchAndDevelopment ] ] .
```

```
ex:policy-assetcollection a odrl:Policy ;
  odrl:uid ex:policy-assetcollection ;
  odrl:profile dpv-odrl;;
  odrl:permission [
    odrl:target [
      a odrl:AssetCollection ;
      odrl:source ex:data-catalog ;
      odrl:refinement [
        odrl:leftOperand dpv-odrl:PersonalData ;
        odrl:operator odrl:isA ;
        odrl:rightOperand pd:HealthRecord ] ] ;
    odrl:assigner ex:data-subject ;
    odrl:action dpv-odrl:Use ;
    odrl:constraint [
      odrl:leftOperand dpv-odrl:LegalBasis ;
      odrl:operator odrl:eq ;
      odrl:rightOperand eu-gdpr:A6-1-a ] ] .
```

Policy Enforcement



Policy Enforcement



Slabbinck, Wout, Julián Rojas Meléndez, Beatriz Esteves, Pieter Colpaert, and Ruben Verborgh. **‘Interoperable Interpretation and Evaluation of ODRL Policies’**. In *The Semantic Web*, edited by Edward Curry, Maribel Acosta, Maria Poveda-Villalón, et al. Springer Nature Switzerland, 2025. https://doi.org/10.1007/978-3-031-94578-6_11.

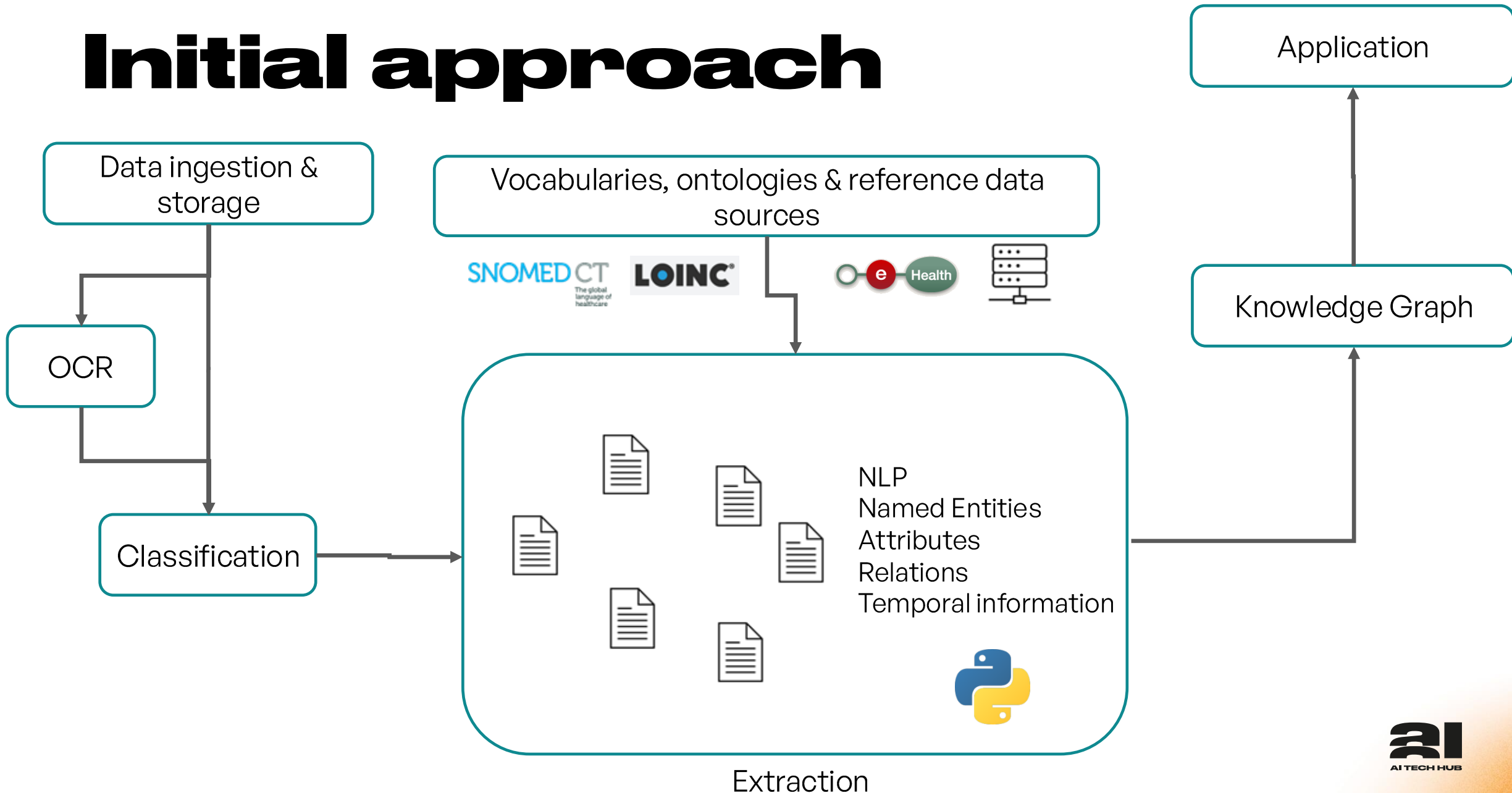
Information Extraction

- How to extract metadata, content and context from existing data?
- Mainly document based data
- Context often missing

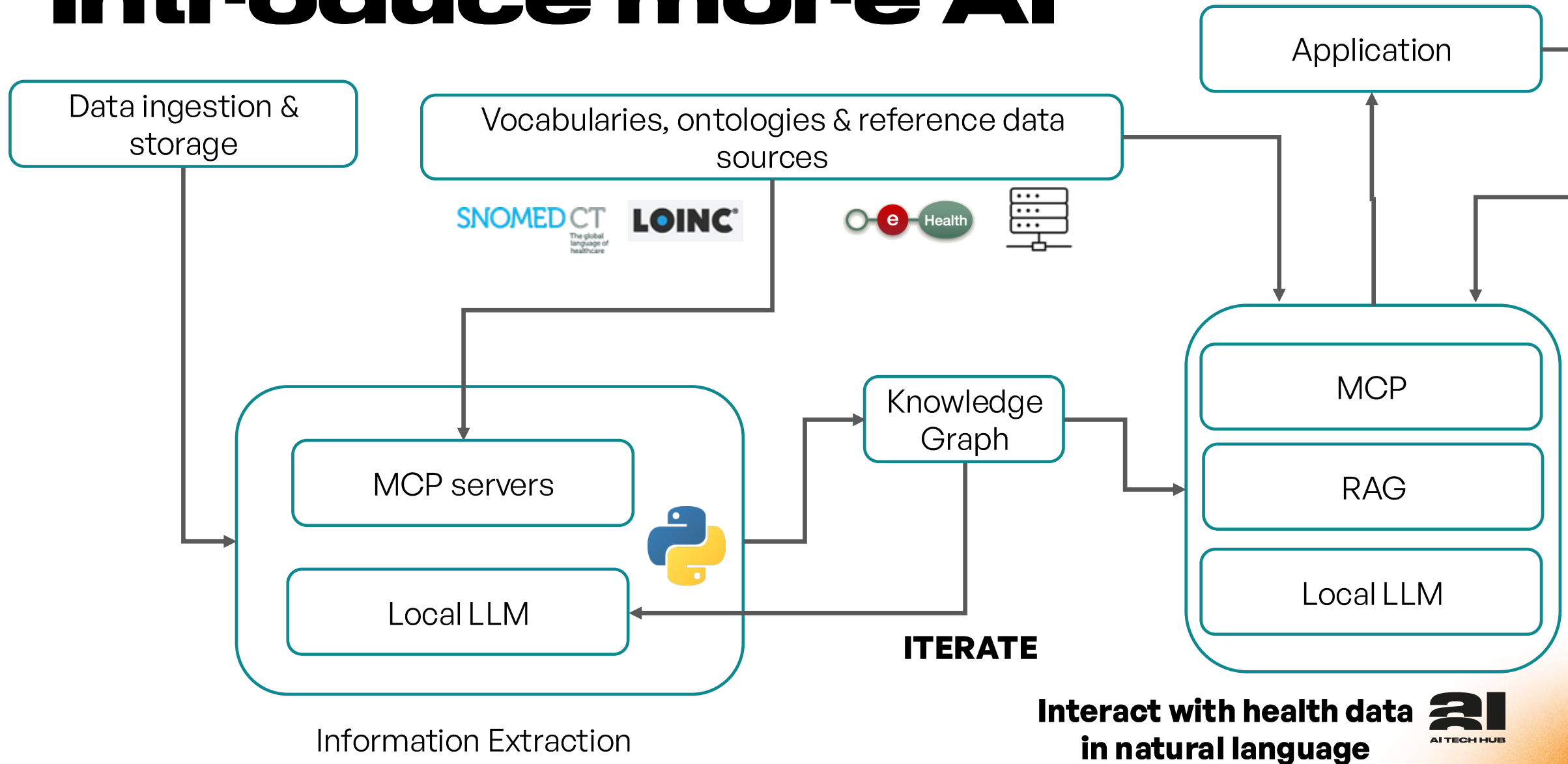
Initial approach

- Classical ETL (Extraction Transformation Loading) approach
- Build a pipeline and try to deal with the complexity
 - Different types of documents
 - Different ways of representing data
 - Missing context
 - Scanned vs pdf-generated data

Initial approach



Introduce more AI



Results

- Quick wins
 - Difficult process with classical approach
- Fast evolution in quality and performance of available tooling
- Building in iterations more straightforward
 - Need training data!
- Crap in > crap out
 - Importance of standards, references, semantics
- Evolution of computation requirements?
- New innovative ways of interacting with data for end-users

Summary & next steps

- Trust is everything
 - Key role for Policy Management in future data & AI applications
- Major technological switch ongoing
 - Step away from 'classical approaches' in information extraction
- Renewed attention for semantics
 - Neuro-symbolic AI

Next steps post-PACSOI

Transition research to production

- Make ODRL+DPV-based Policy Management more industry-proof
- Inclusion in Health Data Vault offering
- Include initial version of new LLM-driven information extraction in end-user application of FAQIR Institute (B2C)
- Include more functionalities in different iterations
 - Interact with health data in natural language
- Expand to high-throughput version (B2B2C)



Thank you!

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THE BRAIN

AI TECH HUB